

Artificial intelligence-based tools with automated segmentation and measurement on CT images to assist accurate and fast diagnosis in acute pancreatitis

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Abstract

Objectives: To develop an artificial intelligence (AI) tool with automated pancreas segmentation and measurement of pancreatic morphological information on CT images to assist improved and faster diagnosis in acute pancreatitis.

Methods: This study retrospectively contained 1124 patients suspected for AP and received non-contrast and enhanced abdominal CT examination between September 2013 and September 2022. Patients were divided into training ($N=688$), validation ($N=145$), testing dataset [$N=291$; $N=104$ for normal pancreas, $N=98$ for AP, $N=89$ for AP complicated with PDAC (AP&PDAC)]. A model based on convolutional neural network (MSAnet) was developed. The pancreas segmentation and measurement were performed via eight open-source models and MSAnet based tools, and the efficacy was evaluated using dice similarity coefficient (DSC) and intersection over union (IoU). The DSC and IoU for patients with different ages were also compared. The outline of tumour and oedema in the AP and were segmented by clustering. The diagnostic efficacy for radiologists with or without the assistance of MSAnet tool in AP and AP&PDAC was evaluated using receiver operation curve and confusion matrix.

Results: Among all models, MSAnet based tool showed best performance on the training and validation dataset, and had high efficacy on testing dataset. The performance was age-affected. With assistance of the AI tool, the diagnosis time was significantly shortened by 26.8% and 32.7% for junior and senior radiologists, respectively. The area under curve (AUC) in diagnosis of AP was improved from 0.91 to 0.96 for junior radiologist and 0.98 to 0.99 for senior radiologist. In AP&PDAC diagnosis, AUC was increased from 0.85 to 0.92 for junior and 0.97 to 0.99 for senior.

Conclusion: MSAnet based tools showed good pancreas segmentation and measurement performance, which help radiologists improve diagnosis efficacy and workflow in both AP and AP with PDAC conditions.

Advances in knowledge: This study developed an AI tool with automated pancreas segmentation and measurement and provided evidence for AI tool assistance in improving the workflow and accuracy of AP diagnosis.

Keywords: artificial intelligence; convolutional neural network; computed tomography; dice similarity coefficient; intersection over union; acute pancreatitis.

Introduction

Acute pancreatitis (AP), as a common digestive emergency, is characterized by acute pancreas inflammation and histological acinar destruction.¹ The annual case number of AP is 120 000–450 000 in the world with an annual incidence increasing from 0.19% to 0.71% in recent 20 years.² Due to the development of local inflammation involving systemic organs, AP may become severe, leading to a fatality rate of 13–35%.¹ Therefore, it is significant to promptly diagnose and predict the progression of AP to prevent the occurrence of complications.

CT has become a primary imaging modality for diagnosing AP. However, the efficacy of CT diagnosis is heavily reliant on the experience and fatigue level of the radiologist, as it largely depends on visual judgement.³ The main CT diagnostic criteria for AP include peri-pancreatic oedema and

pancreatic morphological enlargement.⁴ However, pancreatic edge blur and similar grey values in the peri-pancreas make it difficult to recognize the contour and determine the expansion.^{5–7} Additionally, AP may be complicated by other pancreatic diseases, such as pancreatic ductal adenocarcinoma (PDAC), leading to a missed diagnosis and potential fatality. Given the distinct clinical decisions required for AP and PDAC, it is essential to accurately differentiate between them.^{4,6,7} To address these limitations, automatic intelligence-based grey value to distinguish pancreatic edge blur has been developed, utilizing deep learning and computer vision (CV) to achieve pixel-level segmentation of organ contours. This technology can automatically highlight (oedema, tumours, and pancreas) and measure (the area and length of the maximum section) lesion and organ contours,

assisting radiologists in improving diagnosis efficacy and providing immediate features to assist clinical decision-making.⁸

The accuracy of organ measurement using artificial intelligence (AI)-based segmentation is dependent on the accurate segmentation of the organ.⁹⁻¹¹ Therefore, organ segmentation on CT images has become an important task in recent years, with reported satisfactory accuracy (e.g. Dice similarity coefficient (DSC) > 95%) in the segmentation of organs such as the lung and liver.^{10,12,13} However, pancreatic segmentation remains challenging, and even though methods such as Unet, U2net, and nnUnet have been proposed, their segmentation performance has been relatively poor.¹⁴⁻¹⁶ This is due to several factors, including the discontinuity of the grey value of the edge pixels of the pancreas tissue and the atypical features of the pancreas tissue compared to the surrounding areas.¹⁴ Blurred boundaries in exudative pancreatitis are generally defined by an experienced physician using CT grey values to define blurred boundaries in the pancreatic area (such as parenchyma mixed oedema, oedema mixed fat). Additionally, the poor contour and reduced contrast of the internal pancreatic glands and fat on CT images pose a challenge,^{14,15} as does the blurred boundary of exudative

pancreatitis, which is difficult to define.¹⁷ Further research is needed to develop accurate and robust methods for pancreatic segmentation, which would improve the accuracy of CT diagnosis for pancreatitis and related diseases.

In this study, we developed a convolutional neural network (CNN) based AI tool and compared the segmentation and measurement performance among this tool and a six of open-source tools, then explored the diagnostic efficiency in AP and AP&PDAC when radiologists were assisted by the AI tool.

Methods

Dataset preparation

This retrospective study was approved by the Ethics Committee of Medicine. Written informed consent is waived by the IRB. A total of 2180 patients suspected with AP and received abdominal CT from September 2013 to September 2022 in Hospital were screened. Exclusion criteria: (1) patients with AP complicated by hyperlipidaemia or benign tumour; (2) patients with incomplete clinical information; (3) patients with severe pancreatic atrophy.

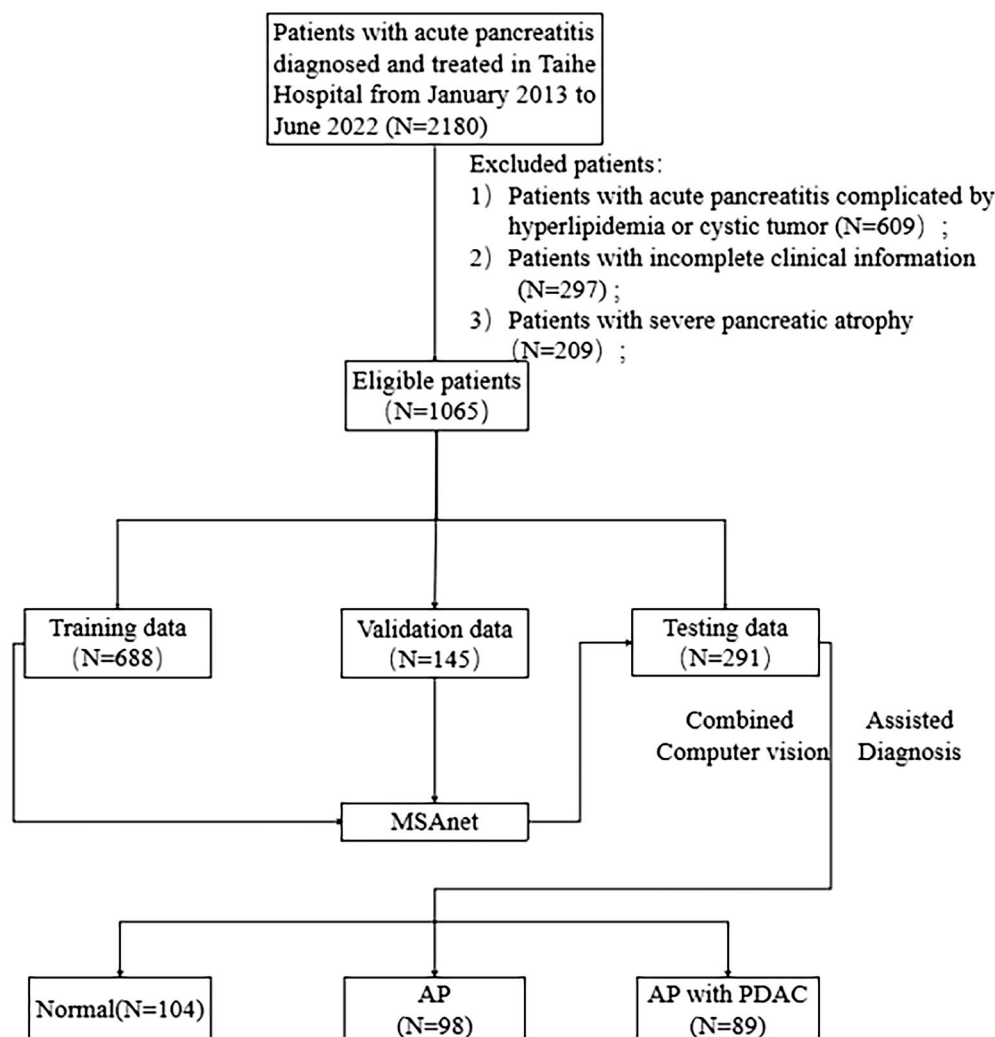


Figure 1. Main flowchart and processing steps.

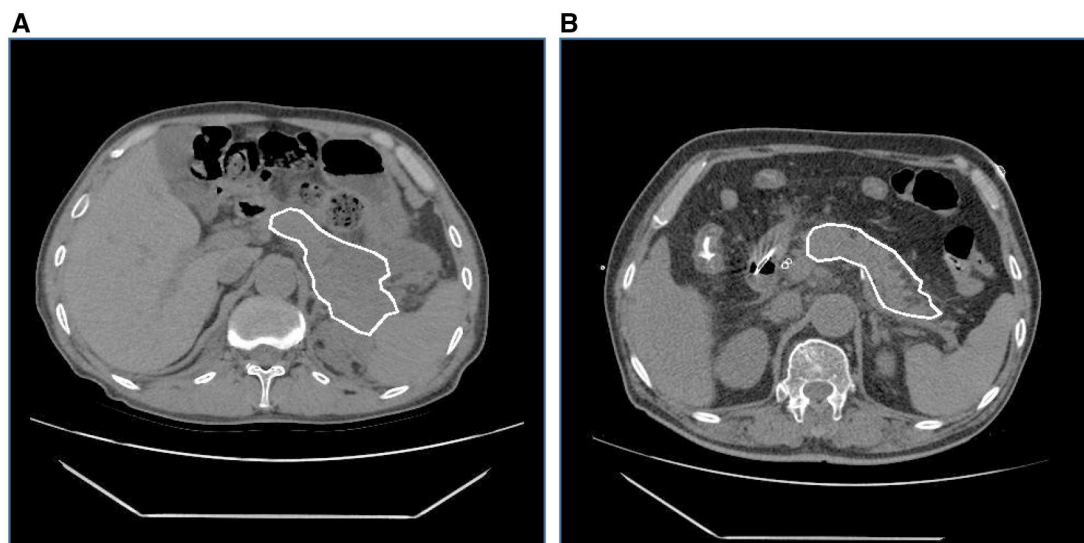


Figure 2. Comparison of segmentation maps generated by radiologists in non-enhanced CT images was conducted for two types of acute pancreatitis: (A) Haemorrhagic necrotic acute pancreatitis and (B) exudative oedema acute pancreatitis.

As shown in Figure 1, this study finally included 1124 patients, according to the ratio of 6:1:2, the patients were divided into training set ($N=688$), validation set ($N=145$), testing set ($N=291$; $N=104$ for normal pancreas, $N=98$ for AP, $N=89$ for AP&PDAC). In addition, to explore the effect of age-related factors (such as lobulation and oedema) on segmentation performance, we first collected all old patients (55-75 years old, $N=90$), then used Python's random seed feature to randomly select 90 patients in the middle-aged (25-54 years old) and young individuals (15-24 years old), respectively.

The radiologic diagnosis (pancreas get enlarged and diffuse tissue fluid in the peri-pancreatic region) conformed the World Health Organization diagnostic criteria for AP from 2007 to 2016. General information and clinical data such as age, gender, height, urinary amylase (the maximum value of urinary amylase at 12-24h after onset), severity, and other histopathological results were obtained from medical records.

CT imaging protocols

All screened patients received abdominal non-contrast CT and contrast-enhanced CT scanning on a 64-row multidetector CT (OPTIMA 660, Milwaukee, WI, USA). The scanning parameters were as follows. Scan range: from the top of diaphragm to the line connecting of the anterior superior iliac spines on both sides, tube voltage: 120 kVp, automatic tube current modulation (noise index = 10, range: 100-350 mA), pitch: 0.984:1, rotation speed: 0.6 s, section thickness: 5 mm, interval: 5 mm, image reconstruction: adaptive statistical iterative reconstruction with blending ratio of 30% at 1.25 mm thickness. Following the pre-contrast enhanced scan, 80-90 mL iodinated contrast agent (Ioversol, Hengrui Medicine, China) was injected intravenously at a flow rate of 2.5-3 mL/s using a high-pressure syringe. The arterial phase scans started with 30 s delay after injection of contrast agent, and the portal venous phase scans with 60 s delay.

Image segmentation

The contours of AP were manually segmented by two junior abdominal radiologists (F.L. and Y.G., with 4~5 years of

experience), and another senior radiologist (L.W., with 25 years of experience) checked the contours. Abdominal radiologists used the medical image annotation tool (LabelMe, <https://github.com/jiangjiawen/labelme-for-dicom>) to mark the entire pancreas (including contrast enhancement and pancreatic parenchyma) on the axial scan layer and were blind to the patient clinical information. If the contour is not clear on non-enhanced CT, the radiologist could mark non-enhanced CT based on location information and CT image continuity information (Figure 2).¹⁷⁻²¹ For intra-observer producibility analysis, the same volumes of interest on CT images of 30 randomly selected patients were segmented twice by a radiologist (F.L.). Inter-observer reproducibility analysis was performed based on the same cohort abovementioned, and another radiologist evaluated the differences between two segmentations (L.W.). Consistency was assessed by calculating the dice similarity coefficient (DSC) and the intersection over union (IoU).

After the model training, validating, and testing, the extra contour fixing by two radiologists was conducted on the testing dataset for extended clinical applications, including benchmarking AI performance, identifying areas for improvement, assessing clinical usability, and continuous learning. Weighting files contain parameters (named weights) that the model learned during the training process, which determine how the model reflects output from the input data, indicating the strength of the neural network. Weighting files enable the model to install on varied environments and devices; thus, the model could execute the preconfigured task *via* the uploaded corresponding weight files. In our study, we used the weighting files to execute segmentation tasks on the validation dataset and the testing dataset. The applicability of AI-generated weighting file was evaluated by two radiology experts. In detail, the AI-generated annotated of 291 patients in testing dataset were assigned to radiologists (X.Y. and J. K., with 11-12 years of experience) and fixed *via* the volume correction magnitude (defined as the quotient of edited volume divided by the AI-generated volume). The accuracy of the classification was determined as follows: (1) No correction required, volume contours above 0-0.2 need to be

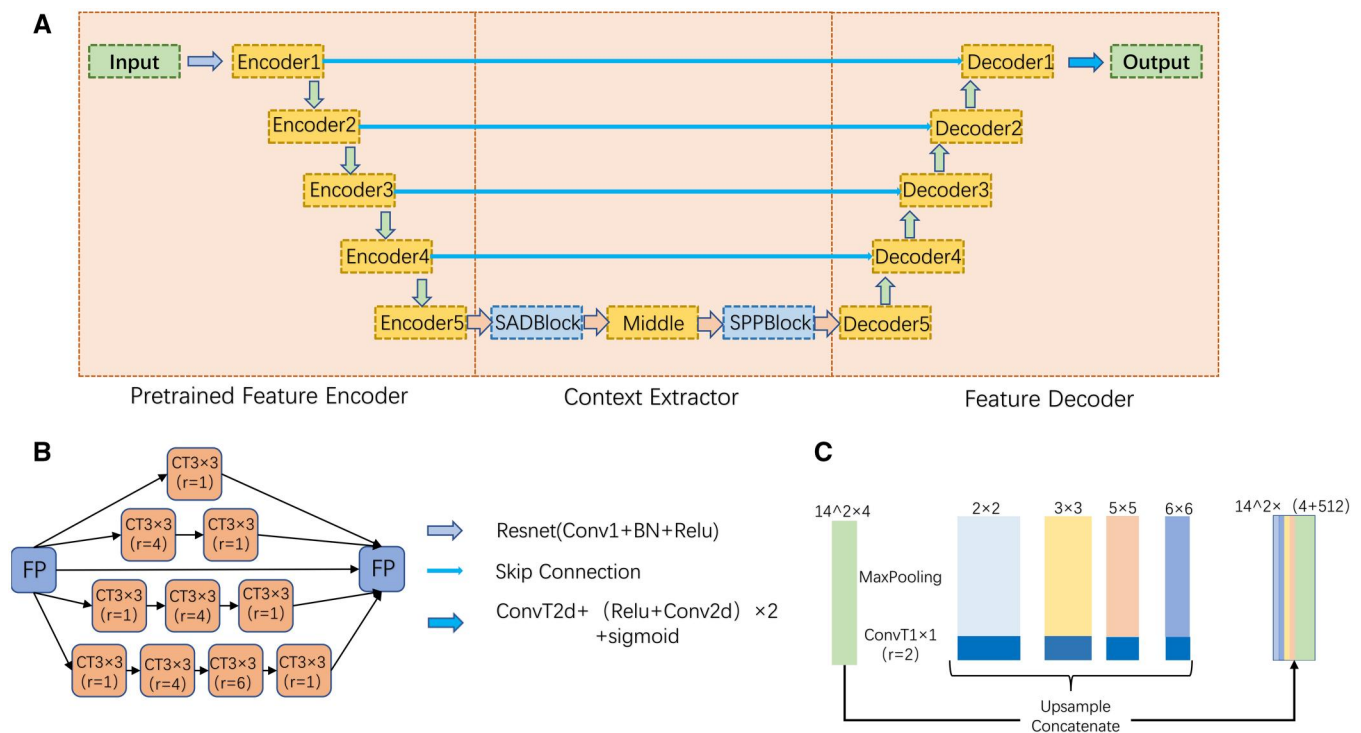


Figure 3. The frame structure and all modules in the MSA model. (A) The main model frame diagram, which is the main framework of Unet (MSAnet). In the Unet architecture, the decoder uses deconvolution (or transposed convolution) layers to perform upsampling of the feature maps. This operation is the inverse of convolution, where the model learns to upsample a feature map to a higher resolution, which helps in reconstructing the segmentation map at the original scale of the image. (B) The multi-scale atrous convolution block (MAC) module SADBBlock (spatial attention block), which mainly obtains the different receptive fields and granularities of features in different scales. (C) The spatial pyramid pooling block (SPP) block, which mainly obtains deep features to avoid deepening the depth of model.

corrected. (2) A volume profile of 0.2-0.4 or more required a volume curve of 0.4-0.6. (3) Volume percentages over 0.6-0.8 need to be modified to 0.8-1 volume curves. (4) And the volume curve that exceeded 0.8 to be modified to the volume curve of 1.

AI pancreas contour tool

A MSAnet model was developed in this study. In view of the excellent performance of Unet in the segmentation of liver, lung, kidney and spleen,⁹⁻¹¹ we selected Unet as basis to segment the pancreas. Although the architecture of Unet helps to preserve the details of the original image, there are some problems about continuous pooling lose the edge information of AP and span convolution reduce the resolution of feature.¹⁴⁻¹⁶

At the same time, we retained the symmetric structure of traditional Unet and added a multi-scale atrous convolution module to synthesize the different receptive fields and granularities of features in medical image so as to make up for the lost information of pancreatic edge caused by continuous pooling. In addition, considering the dilemma between network depth and overfitting, a spatial pyramid convolution module helps learn the depth features and obtain the difference of resolution without a large depth of model.¹⁴ Meanwhile, the pretrained ResNet34 module in MSAnet is used to speed up model convergence, utilize the function of relu and dropout to avoid overfitting.^{15,16}

Hence, the developed MSAnet was mainly divided into three parts: pretrained feature encoder, context extractor, and feature decoder. In the encoder, we use the pre-trained Resnet module to replace the traditional Unet module, which could speed up the model convergence speed. In the context

extractor, we added a multi-scale atrous convolution block (MACBlock) to compensate the missing information caused by continuous pooling, and we hoped the spatial pyramid pooling block (SPPBlock) could learn depth feature on CT images without deepening the depth of network (as shown in Figure 3). In the feature decoder, this module kept the basic architecture for the decoder of Unet to learn the factor of decoding in the context extractor. Compared with ordinary convolution, atrous convolution could obtain a larger receptive field when recognized small target classification tasks such as pancreas to obtain denser data and preserve the spatial characteristics of the image without the missing information of image.^{11,16}

At the same time, we also used open-source Unet, AttUnet, U2net, DconnNet, SDC-UDA, etc. as comparisons to evaluate the segmentation performance of developed MSAnet, and to discover potential segmentation models to improve the performance of pancreatic contours (Figure 3). The programs were written through an open-source named Pytorch 1.4.1 on Python 3.8. The input images were loaded on RTX3060 [Ubuntu 14.04; 3060Ti (16GB)] for training. The machine had 11th Gen Intel i7-11700, 2.50 GHz processor, 32.0GB of DDRAM and NVIDIA GeForce with CUDA 11.6.0.

AP is a sudden inflammation of pancreas. The common causes are gall stone, alcohol abuse, systematic disease, trauma, etc. In AP pancreas get enlarged and diffuse tissue fluid in the peri-pancreatic region (PR) (peri-pancreatic oedema). In patient with suspected AP, the radiologist should have clear information about intra pancreas region (IR) (enlargement of pancreas, low density lesion) and PR (diffuse oedema) to assist the diagnosis.^{17,22,23} Although the IR was obtained through deep learning

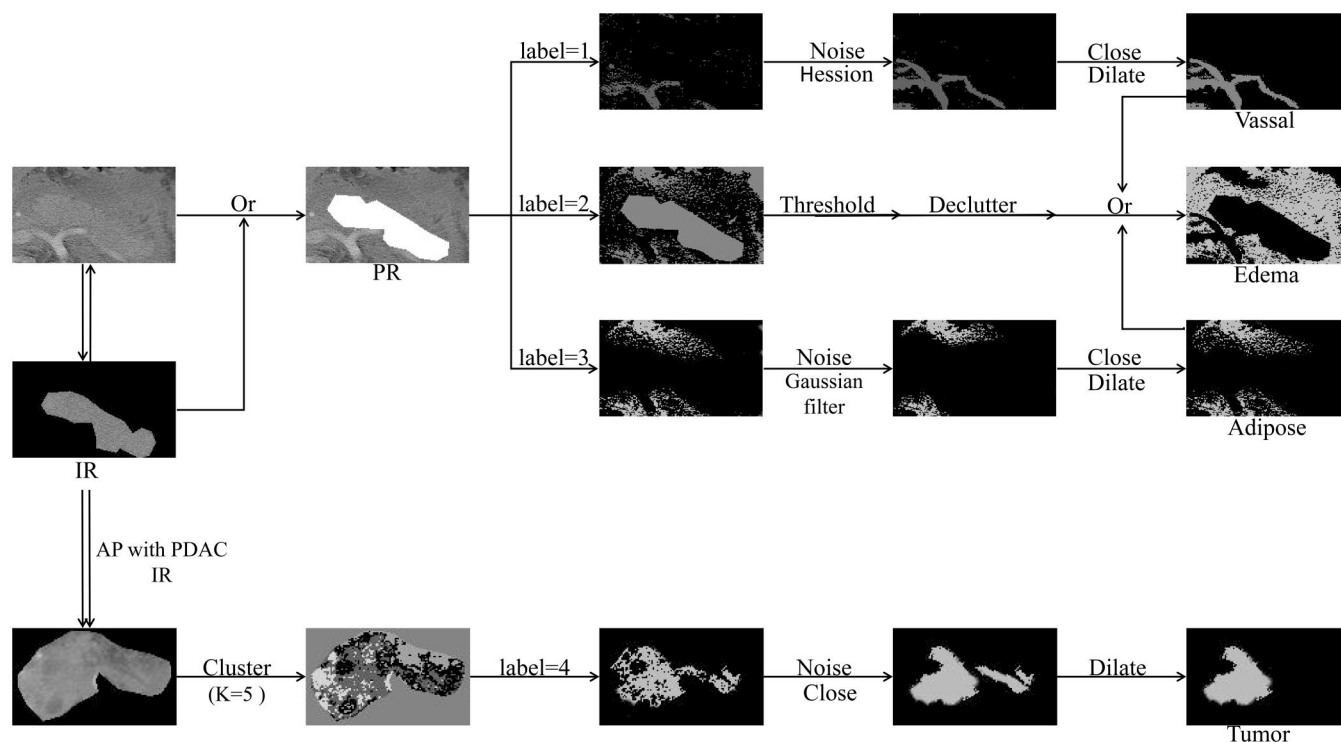


Figure 4. The process of obtaining the oedema area and tumour area in a CT image through computer vision involves several steps.

technique, it is necessary for radiologist to diagnose peri-pancreatic oedema. The approximate elliptical contour and length and width of pancreas were obtained as well as the key points were obtained. Based on the key point, we increased the width by 1.5 times and slightly expanded the length by 1.1 times to maximize the removal of disturbing information and obtain the pancreatic profile, containing PR and IR information. The PR was obtained by OR calculated based on the IR contour and pancreatic contour. *K*-means clustering was used to extract IR's tumours and PR's oedema, based on the image of TR and PR. In suspected AP, IR image of most of them were clear pancreatic parenchyma, showed low density lesion in AP and PDAC. Although the outline of the tumour is blurred for human eyes, it is easier to treat the IR's tumour based on the clustering algorithm. For suspected AP of PR image, oedema is a mixture of fat and blood vessels, the edges are blurred, and the grey value is close, that leads a challenge to extract the oedema region by clustering (*K*-means). Therefore, we used clustering and CV methods (including methods like dilation, erosion, clustering, etc.) to obtain the contours of blood vessels and fat first and subtracted the contours containing oedema to obtain the true oedema contour morphology (Figure 4).

Diagnosis efficiency with assistance of AI tool

The diagnosis efficiency evaluation was performed on the testing dataset (dataset 1: AP, $n = 98$; dataset 2: AP&PDAC, $n = 89$; dataset 3: normal, $n = 104$). Firstly, we anonymized and randomized the data from 291 patients and transferred the images to two radiologists for diagnosis. A junior radiologist (W.K., with 5 years of experience) and a senior radiologist (P.Z., with 15 years of experience) performed independent diagnosis with and without tool assistance, respectively. The tool-assisted diagnosis was performed based on the MSAnet and CV to segmented pancreas images and automated generated pancreas measurement. AI tools would

provide the segmented pancreas, measurements including area, length, wideness, height, as well as the vessel, fat, oedema structure derived from clustering and CV, as shown in Figure 6. The results of the diagnosis were compared between the two conditions: using AI tools and manual diagnosis. To eliminate bias, the radiologists were blind to the patient information and reference results, and the MSAnet tool-assisted diagnosis was conducted at 2 weeks after non-assisted diagnosis. The diagnosis time and results were recorded.

Statistical analysis

Clinical data comparison

IBM SPSS version 22.0 was used for clinical data analysis. Category data were shown as number (percentage), continuous and normally distributed data were displayed as mean $\pm$ standard derivation ($\bar{x} \pm s$). The gender ratio comparison between training and validation dataset was performed using Chi-square test. Other clinical data comparison was conducted using independent sample *T* test. For the comparison among three age subgroups, one-way analysis of variance method was utilized. $P < .05$ was considered as statistically significant.

Segmentation efficacy analysis

DSC and IoU were used to evaluate the performance of segmentation.¹⁶ There are two slightly different measures of similarity between ground truth and algorithmic predictions

$$DSC(P, G) = 2|P \cap G| / (|P| + |G|)$$

$$IoU(P, G) = 2|P \cap G| / |P \cup G|$$

In the formula, *P* delineated by abdominal radiologist is the real pancreas contour area, *G* is the predicted range of pancreas contour segmented by models. Using IoU as the loss

of model is better than using DSC if high prediction efficacy and large overlap area (the prediction box needs to coincide with the ground truth as much as possible) are required. However, it is possible that the higher of prediction similarity, the higher of indicator oscillation.^{11,16} Thus, adopting both DSC and IoU as an evaluation index would more reasonably evaluate the performance of model segmentation.

Diagnosis efficacy analysis

For diagnosis efficiency evaluation, SPSS software was only employed for picturing receiver operating characteristic (ROC) curves and calculating the area under curve (AUC). In our study, there were three categories (normal, AP, AP&PDAC). To perform ROC analysis, we combined two categories into one new category for the detection of the other status. For normal detection, AP and AP&PDAC patients were combined as negative; for AP detection, normal and AP&PDAC patients were combined as negative; and for AP&PDAC detection, we combined AP and normal as negative. Referring to the previous studies, we calculated the sensitivity and specificity based on the reference and radiologist diagnosis results using Python 3.8.0_sklern.

Results

Patient baseline

Table 1 and Table S1 summarized patient characteristics. There were no statistically significant differences among the training, validation, and testing datasets in terms of disease category, age, gender, height, weight, BMI, pancreatic lipase, blood glucose, WBC, RBC, morphological evaluation (all $P > .05$). In addition, patients with different ages showed no statistical differences in disease category (frequency of AP&PDAC), pancreatic lipase and blood glucose (all $P > .05$) (Table S1).

Performance of MSAnet based tool

The accuracy of pancreatic profiles generated by different models is summarized in Table 2. Among all model-based tools, MSAnet had the highest DSC of 0.966 and the second highest IoU of 0.953 (the highest was 0.954) on the training

dataset, the highest DSC of 0.819, and the third highest IoU of 0.706 on the validation dataset. Figure 5 exhibits the difference of pancreatic segmentation effect between different periods.

In a module ablation experiments (Table S2), when retaining the SAD module, the DSC and IoU on the validation dataset were decreased from 0.819 to 0.799 and 0.706 to 0.670, respectively. While only preserving the SPP module, the DSC and IoU were reduced to 0.785 and 0.662, respectively. When removing both modules, the DSC and IoU were respectively further decreased to 0.777 and 0.651.

MSAnet based tool also achieved good performance in AP&PDAC [mean DSC: 0.817, max: 0.821, min: 0.809, CI: (0.811, 0.819), mean IoU: 0.756, max: 0.761, min: 0.749, CI: (0.751, 0.759)], simple AP [mean DSC: 0.823, max: 0.829, min: 0.817, CI: (0.819, 0.826), mean IoU: 0.767, max: 0.772, min: 0.760, CI: (0.761, 0.769)], and normal pancreas [mean DSC: 0.832, max: 0.836, min: 0.825, CI: (0.827, 0.835), mean IoU: 0.778, max: 0.782, min: 0.771, CI: (0.773, 0.780)]. As shown in Figure 6, with MSAnet based tool utilized, for CT images on different contrast enhanced phases, the contour of pancreas could be automatically delineated and the area, length, wideness, and height of pancreas would also be simultaneously evaluated enlarged pancreas and shown lesion for radiologists.

The performance of MSAnet could be influenced by the age of patients. Therein, higher the age, lower the DSC and IoU. In detail, the segmentation efficacy on old group was worst with a DSC of 0.946 and a IoU of 0.932, while young group had best performance (DSC: 0.968, IoU: 0.954), and middle group showed middle efficacy with DSC of 0.951 and IoU of 0.948.

Efficacy comparison between MSAnet based tools assisted diagnosis and manual diagnosis

Table 3 and Figure 7 compare the diagnostic performance of manual diagnosis and AI-assisted diagnosis. With the assistance of MSAnet based tool, the diagnosis time for junior radiologist was reduced by 26.8%, which was reduced by 32.7% for senior radiologist. Junior radiologist had significantly improved diagnostic efficiency by MSAnet assistance:

Table 1. Patient characteristics in datasets.

Patient characteristics	Training dataset	Validation dataset	Testing dataset	P-value
Total number of cases	688	145	291	
AP with PDAC [<i>n</i> (%)]	210 (30.5)	44 (30.4)	89 (30.5)	.051
AP [<i>n</i> (%)]	229(33.4)	48(33.1)	98(33.6)	.056
Female gender [<i>n</i> (%)]	193 (36.1)	51 (35.1)	99 (34.0)	.064
Age ($\bar{x} \pm s$, years)	46 $\pm$ 8.36	43 $\pm$ 6.36	41 $\pm$ 5.36	.053
Height ($\bar{x} \pm s$, cm)	167 $\pm$ 8.72	164 $\pm$ 8.43	168 $\pm$ 6.42	.061
Weight ($\bar{x} \pm s$, kg)	61.3 $\pm$ 20.3	59.7 $\pm$ 19.5	58.4 $\pm$ 22.3	.125
BMI ($\bar{x} \pm s$)	24 $\pm$ 7.4	21 $\pm$ 6.1	22 $\pm$ 5.4	.221
Biochemical indicators				
Pancreatic lipase ($\bar{x} \pm s$, u/L)	195 $\pm$ 32.31	200 $\pm$ 30.31	197 $\pm$ 25.61	.483
Blood glucose ($\bar{x} \pm s$, Mmol/L)	9.3 $\pm$ 2.34	9.1 $\pm$ 1.12	9.4 $\pm$ 0.92	.214
Blood routine index				
RBC ($\bar{x} \pm s$, T/L)	3.1 $\pm$ 2.1	3.2 $\pm$ 1.8	3.0 $\pm$ 1.9	.089
WBC ($\bar{x} \pm s$, G/L)	7.1 $\pm$ 2.8	6.8 $\pm$ 2.4	7.2 $\pm$ 2.3	.221
Imaging score				
Morphological evaluation ($\bar{x} \pm s$)	2 $\pm$ 0.4	2 $\pm$ 0.3	2 $\pm$ 0.2	.619

Table 2. The performances of different models in pancreas segmentation.

Segmentation model	Training dataset		Validation dataset	
	DSC	IoU	DSC	IoU
MSAnet	0.966	0.953	0.819	0.706
Attnet	0.962	0.950	0.799	0.670
Unet	0.957	0.938	0.785	0.662
Res34Unet	0.963	0.948	0.777	0.651
FCNUnet	0.962	0.951	0.791	0.701
U2net	0.958	0.954	0.781	0.651
Channel Unet	0.933	0.923	0.797	0.662
DconnNet	0.956	0.946	0.791	0.732
SDC-UDA	0.960	0.951	0.804	0.762
Mean loss	0.957	0.946	0.794	0.763

Abbreviations: DSC = dice similarity coefficient; IoU = intersection over union.

in normal, AUC increased from 0.97 to 1, specificity increased from 94.2% to 100%; in AP, AUC increased from 0.91 to 0.96, specificity increased from 84.4% to 92.2%; in AP&PDAC, AUC increased from 0.85 to 0.92, specificity increased from 73.0% to 82.1%, in sum, the accuracy increased from 89.6% to 92.9%, time reduced from 106 ± 42 to 81 ± 37 sec. Senior radiologist had decreased diagnostic time for AP and diagnostic efficiency for AP with PDAC by using MSAnet: in normal, AUC increased from 0.99 to 1, specificity increased from 99.3% to 100%; in AP, AUC increased from 0.98 to 0.99, specificity increased from 93.7% to 98.9%; in AP&PDAC, AUC increased from 0.97 to 0.99, specificity increased from 87.6% to 97.7%, in sum, the accuracy increased from 95.8% to 99.3%, time reduced from 76 ± 31 to 52 ± 21 sec.

Discussion

AP is a common condition in radiology emergency department, and CT is one of the most efficient imaging tools. However, due to the unclear contours of pancreas in AP and the possible combination of other diseases such as PDAC, it is prone to misdiagnosis or missed diagnosis.^{1,2,4} Therefore, pancreas segmentation and AI-assisted diagnosis are of great significance.^{10,18} Many previous researches devote efforts into precise segmentation of the pancreas for pancreatic tumour data, whose contours are clearer than those of the AP, so these existing models may not be applicable for the more difficult tasks of AP segmentation and complex clinical situations such as AP&PDAC.^{10,14,15,19} In addition, in the process of AP diagnosis, the change in the pancreas contour size is an important indicator.^{20,21} However, in routine diagnosis, this change is usually determined empirically by radiologists, which faces inevitable subjective bias.^{22,23} Therefore, we established a new deep learning CNN-based tool MSAnet, that contains a pancreas segmentation model and measurement software to segment pancreas and measure contour morphological information. In addition, compared with previous pancreatic segmentation studies which only include typical tumour patients, our study included common AP patients and AP&PDAC patients, and explored the performance of the new tool in various clinical situations of AP and its value for radiologists in assistant diagnosis.

MSAnet based tool could automatically segment and measure the pancreas on CT images in normal pancreas, simple

AP and AP complicated PDAC with good efficacy, showing best performance among seven models. In the diagnosis of AP and AP&PDAC, with the assistance of MSAnet tool, not only the time spent was shorten, but also the efficacy was improved.

In the cross-sectional study, MSAnet achieved the best results among nine including Unet, DconnNet, SDC-UDA, et al. It is profited from muti-scale module and symmetrical structure co-constructing MSAnet model, which preserved the details of the image while dealt with the problems of unclear edges of the pancreas and the close grey scale values of the tissue.^{21,23,24} In the modules ablation experiment (Table S2), the SPP module and SAD module significantly enhance the segmentation performance of the model. Studies have shown that models with similar structures containing multi-scale modules and symmetrical structures are suitable for medical segmentation tasks dealing with smaller organs and detailed lesions, such as better performance in retinal blood vessel detection, cell contour segmentation, and retinal OCT layer segmentation (DSC = 0.958 ± 0.003, IoU = 0.948 ± 0.003 in DRIVE Dataset).²⁵ In comparison to these studies, this MSAnet exhibited higher performance (IoU = 0.953, DSC = 0.966 for MSAnet), which may be derived from different scale parameters, structures, and layers, as well as different tasks dealing with. The efficacy differences according to ages could be explained by the physiological process that pancreatic cells undergo necrosis with age and pancreatic atrophy caused by fat cells filling the pancreatic tissue, which will make it more difficult to segment the pancreas.^{13,26} In addition, the decrease in the grey value of the tail of the pancreas, triggered by pancreatic tail oedema in pancreatitis, could also cause certain analysis errors.^{27,28} The inter-observer and intra-observer reproducibility on pancreas contour segmentation was excellent (intra-observer correlation DSC: 0.978 ± 0.016, IoU: 0.975 ± 0.017; inter-observer correlation DSC: 0.981 ± 0.017, IoU: 0.976 ± 0.015), which not only demonstrates the model's reliability but also emphasizes its validity, as preference and other biases between radiologists have been eliminated.

MSAnet tools provided accurately segmented pancreas contour, clear oedema boundaries (PR), and tumour boundaries (IR) and quantitative morphologic measurements (diameters, perimeter, area) instead of conventional manual pancreas delineation and measurement with inevitable bias, supporting rapid and accurate diagnosis for radiologists. With the assistance of MSAnet based tool, the diagnosis time was decreased and diagnostic efficiency was improved, facing with normal, AP, and AP & PDAC. We also found that junior radiologist (AP&PDAC, AUC = 0.92) using AI tool could approach efficacy close to the manual diagnosis level of senior radiologist (AP&PDAC, AUC = 0.97), and senior radiologist using AI tool could diagnose AP more quickly (32.7% reduction in time) and more accurately (AUC from 0.97 to 0.99) even for difficult task AP & PDAC. Currently, the accurate diagnosis of AP and comorbid conditions such as AP&PDAC are very dependent on the level of experiences and mental state of the imaging doctor.^{3,29} Thus, in the fact of the high patient load in emergency radiology departments in some regions, AI tools demonstrated their values.^{4,30,31} With the help of MSAnet based tools, it allows rapid diagnosis and more cases diagnosis in a limited time with optimized workflow; additionally, it make AP diagnosis easier and might release the mental pressure for doctors. Collectively, this

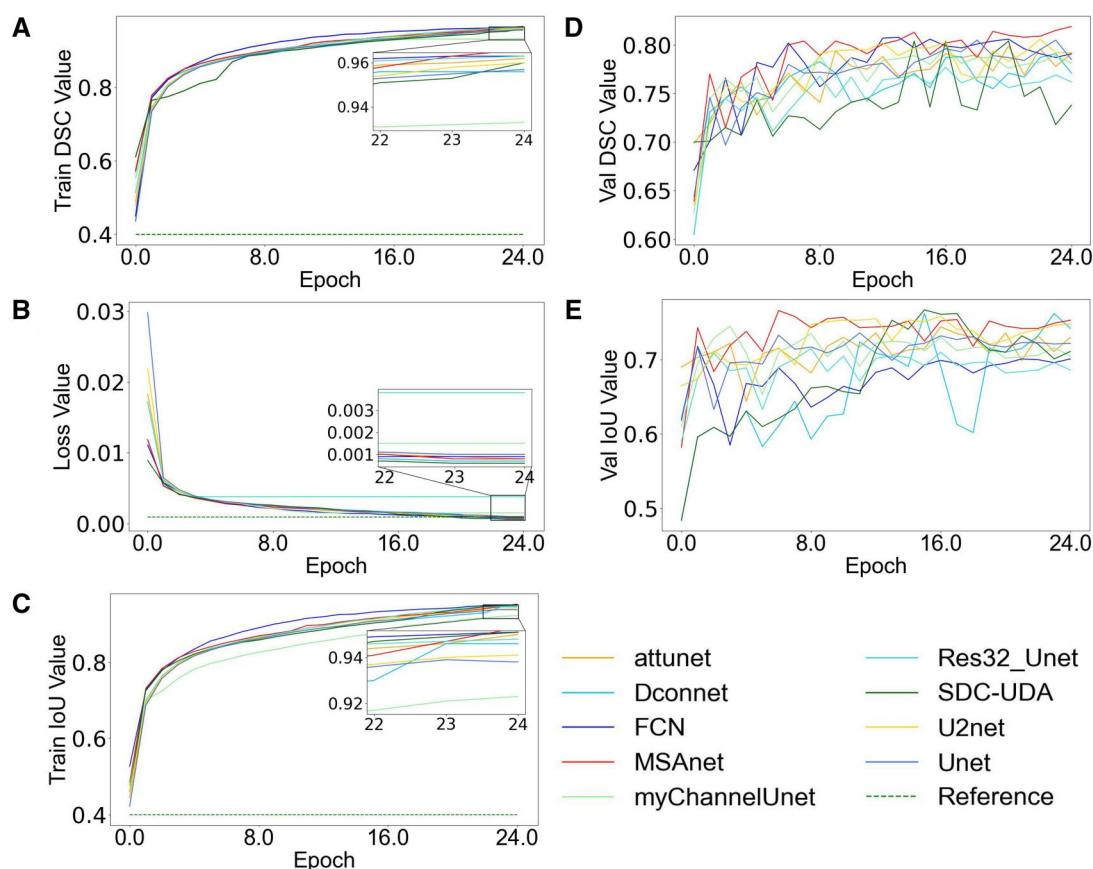


Figure 5. Performance of each sub-network in the training and validation datasets. MSAnet had the best effect. (A) Convergence of DSC in the training set. (B) Convergence of loss. (C) Convergence of IoU in the training set. (D) The change of DSC in validation set. (E) Convergence of IoU in the validation set.

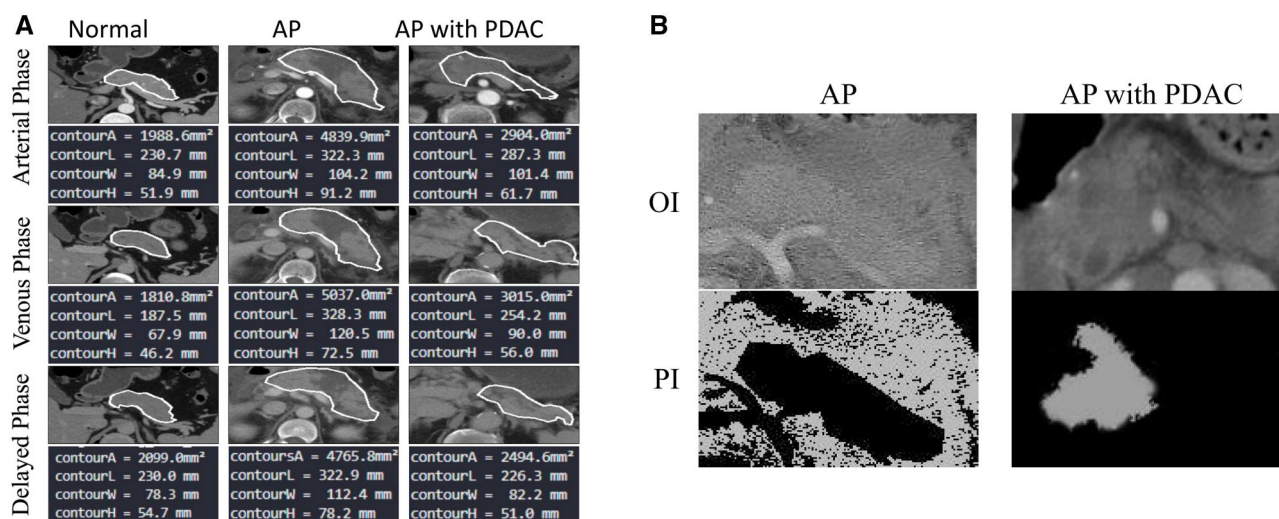


Figure 6. Performance of AI tool in segmentation map (pancreas, oedema, and tumour regions) and contour measurements. (A) Comparison of AI tool generated segmentation maps and contour measurements for suspected AP patients in different contrast-enhanced phases; (B) Comparison of AI tool generated segmentation maps for AP and AP with PDAC patients.

MSAnet based tool could mostly benefit patients and doctors in regions with scarce medical resources where doctors are not all sufficiently experienced.

It is undeniable that this study also has some limitations: First, this is a single-centre study, which only used the

datasets from normal, AP, and AP & PDAC to better validate the performance of MSAnet, subsequent multi-centre dataset in pancreatic cyst and benign tumours should be included.^{30,31} Second, the existing imaging methods cannot distinguish between oedema and tumour mixed pancreatic

parenchyma. The clear outline of oedema and tumour obtained by AI is slightly larger than the actual outlines, which can only be used to assist doctor in rapid pancreatic diseases but cannot be used in pancreatic surgery.^{17,30} Third, it is infrequent and challenging for AP&PDAC, which could cause the data number of AP&PDAC was smaller than AP in three datasets.^{2,17,32} As a result, the segmentation of testing datasets 1 and 3 slightly lower than testing data 2 and limited sample size in the age error experiment. In the future, we would continue to collect AP&PDAC to balance the errors in the three testing datasets. Moreover, we did not consider the difference of

diagnosing AP and AP&PDAC in the same seniority radiologist. Subsequently, auxiliary performance will be evaluated by adopting more results from the radiologists.³³

In summary, the MSAnet based tool could segment and measure the pancreas on CT images for normal, AP, and AP&PDAC, with age-effected and good performance. Moreover, compared to manual diagnosis, regardless of experience levels of radiologists, with the assistance of this tool, the time spent on reading films was shortened by 32.7%-26.8% but diagnosis accuracy was improved by 3.1%-3.5% for differentiating normal, simple AP, and AP&PDAC.

Table 3. The auxiliary diagnostic efficacy of AI contour tool on assisting radiologists.

	JD	JD_AI	SD	SD_AI
NOM				
AUC	0.97	1	0.99	1
Sensitivity (%)	100	100	100	100
Specificity (%)	94.2	100	99.3	100
AP				
AUC	0.91	0.96	0.98	0.99
Sensitivity (%)	100	100	100	100
Specificity (%)	84.4	92.2	93.7	98.9
AP&PDAC				
AUC	0.85	0.92	0.97	0.99
Sensitivity (%)	100	100	100	100
Specificity (%)	73.0	83.1	87.6	97.7
Sum				
Accuracy (%)	89.6	92.9	95.8	99.3
Time (s)	106 ± 42	81 ± 37	76 ± 31	51 ± 21

Abbreviations: JD = junior doctor; JD_AI = junior doctor + AI; SD = senior doctor; SD_AI = senior doctor + AI.

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Author contributions

Wen Chen and Bo Yang have the same contribution to this article.

Supplementary material

Supplementary material is available at BJR online.

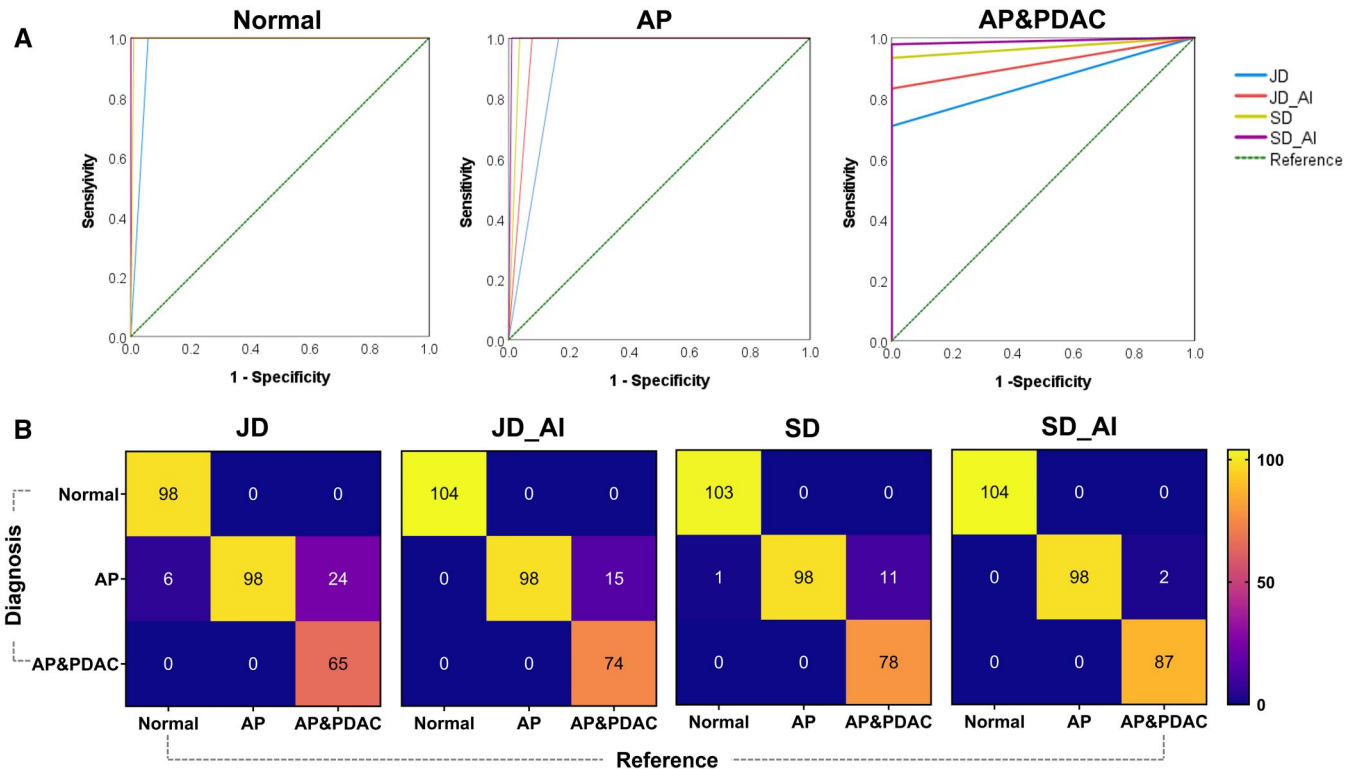


Figure 7. Diagnosis results of suspected AP patients determined by a junior and a senior radiologist with or without MSAnet assistance. (A) ROC analysis; (B) confusion matrix maps. JD = junior radiologist; JD_AI = junior radiologist with the assistance of AI tool; SD = senior radiologist; SD_AI = senior radiologist with the assistance of AI tool.

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Conflicts of interest

None declared.

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